

Publishing a VS Code Project to the Internet Using Ngrok

Technical Experiment Detailed Report

Objective

The purpose of this experiment is to run a Python application locally in Visual Studio Code and make it accessible over the internet using Ngrok. This allows users on other devices and networks to access the application through a secure public URL.

Introduction

When a web application runs on localhost, it can only be accessed from the same computer. Ngrok creates a secure tunnel between the internet and the local machine, enabling remote access without requiring router configuration or public hosting.

Software and Tools Required

- Python Programming Language
- Visual Studio Code (VS Code)
- Ngrok Account and Client
- Internet Connection
- Web Browser

Step-by-Step Instructions

Step 1: Create a Python Application

Create a file named `app.py`. The program starts a simple HTTP server on port 8000. Save the file in your project folder.

Step 2: Execute the Python Program

Open the integrated terminal in VS Code and run the application. After execution, the terminal displays a localhost URL. Open the URL in a browser to verify that the application is running correctly on the local system.

Step 3: Verify Local Access

Before exposing the application to the internet, ensure that the website or application is functioning properly on localhost. Any issues should be resolved before proceeding.

Step 4: Download and Install Ngrok

Visit the Ngrok website, download the appropriate version for your operating system, extract the files, and place the executable in a convenient directory.

Step 5: Create and Configure an Ngrok Account

Register for a free Ngrok account. After signing in, obtain the authentication token from the dashboard and configure it on your computer using the provided command.

Step 6: Start an Ngrok Tunnel

Open a new terminal window and create a tunnel to port 8000. Ngrok generates both HTTP and HTTPS forwarding URLs that connect external users to the local application.

Step 7: Test Public Accessibility

Copy the generated HTTPS URL and open it in another browser, laptop, or mobile device. Verify that the application loads successfully outside the local machine.

Working Process

The Python server listens on port 8000. Ngrok establishes a secure tunnel from the internet to this local port. Incoming requests received by Ngrok are forwarded to the local application, and responses are sent back to the remote user.

Result

The Python application running on localhost was successfully exposed to the internet using Ngrok. Remote users were able to access the application through the generated public URL.